

Mid-Project Submission Report

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Problem Statement

Toxic and hateful content has grown to be a major problem due to the quick expansion of social media and online communication platforms. Offensive language, hate speech, harassment and stereotypes that target people or groups based on identity, gender, race, religion or other characteristics are examples of toxic speech.

The primary way of conventional toxic speech detection techniques is to detect explicit hate speech (stuff like blatantly offensive words or slurs). However, because implicit toxic speech may employ sarcasm, stereotypes, coded language or indirect references that do not contain explicit offensive words, it is more challenging to identify. Models that are able to comprehend semantics and context rather than just keywords are needed to detect implicit toxicity.

Literature Survey

Toxic speech detection is an important Natural Language Processing task due to the increasing amount of harmful content on social media platforms. Early approaches relied on rule based and keyword filtering methods, but these methods were limited because they could not detect implicit hate speech or understand context.

Later, traditional machine learning models such as Logistic Regression and Support Vector Machines were used with text features like Bag-of-Words and TF-IDF. With the advancement of deep learning, models such as CNN, LSTM, and transformer-based models like BERT, RoBERTa, and ELECTRA significantly improved performance by capturing contextual and semantic information. Specifically in this domain, HateBERT [2] which pretrains BERT on a hate speech corpus, has been developed as a strong baseline model for hate speech detection tasks.

Recent research such as the ToXCL [1] framework extends toxic speech detection by combining classification and explanation generation using encoder-decoder architectures.

Description of the Dataset

We have used 2 datasets to train our baselines - IHC [3] and SBIC [4]. In addition, another dataset - HateXplain [5] - was used in the ToXCL framework. Implicit Hate Corpus (IHC) focuses on implicit hate speech and includes explanatory signals, while Social Bias Inference Corpus (SBIC) captures implied social bias and stereotyping in text.

Dataset	Files
IHC (Implicit Hate Corpus)	IHC_train.csv IHC_valid.csv
SBIC (Social Bias Inference Corpus)	SBIC_train.csv SBIC_valid.csv SBIC_test.csv
HateXplain (used to train the Target Generator TG)	TG_train.csv TG_valid.csv

IHC and SBIC are used for toxic speech classification and explanation generation tasks. TG is used for training the target group generator in the ToXCL framework (and was not required/used for the two baselines)

IHC & SBIC	
Column Name	Description
raw_text	Original input text
text	Input text with target group information Format: "Target: {TG} Post: {raw_text}"
explanations	Explanation describing why the text is toxic or biased
output	Combined output containing class label and explanation Format: "class <SEP> explanation"
TG	
Column Name	Description
id	Unique id for each sample
raw_text	Original text post
label	Target group label
target_groups	Target group mentioned in the text

Baselines

Encoder Baseline

The encoder baseline model was used for toxic speech classification. The encoder takes text as input and predicts whether the text is toxic or non-toxic. Multiple pretrained transformer models were used as encoder baselines (HateBERT, BERT-base, ELECTRA-base, and RoBERTa-base). Each model was fine-tuned separately on two datasets (IHC/SBIC), resulting in a total of eight encoder experiments.

The performance of the encoder models was evaluated using accuracy, precision, recall, and F1-score. Additionally, the evaluation loss and the epoch at which the best validation performance was achieved were recorded for each model.

Decoder Baseline

The decoder baseline model was used for text generation tasks such as explanation generation and output text generation. The decoder models were evaluated using different input (text/raw text) and output (output/explanation) column combinations across both datasets (IHC/SBIC), resulting in a total of eight decoder experiments.

The decoder models were evaluated using perplexity as the evaluation metric

Results

Encoder Baseline Results

Dataset Name	Model Name	Eval Accuracy	Eval F1	Eval Precision	Eval Recall	Eval Loss	Epoch
IHCI	HateBERT	0.7904	0.7547	0.7604	0.7501	0.5136	2.0450
IHCI	BERT-base	0.8044	0.7717	0.7769	0.7674	0.5176	2.0450
IHCI	ELECTRA	0.7952	0.7688	0.7650	0.7735	0.4528	2.5562
IHCI	RoBERTa-base	0.8009	0.7722	0.7713	0.7732	0.5158	3.0675
SBIC	HateBERT	0.9008	0.9005	0.9001	0.9014	0.8881	8.0556
SBIC	BERT-base	0.8966	0.8963	0.8960	0.8966	0.9417	8.8889
SBIC	ELECTRA	0.8992	0.8990	0.8987	0.9000	0.5166	4.1667
SBIC	RoBERTa-base	0.8948	0.8944	0.8943	0.8945	0.4150	3.8889

Decoder Baseline Results

Dataset	Text Column	Summary Column	Perplexity (rounded to 4 points)
IHC	text	explanations	38.6669
IHC	text	output	23.3489
IHC	raw_text	explanations	82.4303
IHC	raw_text	output	39.8746
SBIC	text	explanations	23.4471
SBIC	text	output	16.3775
SBIC	raw_text	explanations	37.7867
SBIC	raw_text	output	23.8659

References

1. [Hoang et al. \(2024\)](#) - ToXCL: A Unified Framework for Toxic Speech Detection and Explanation.
2. [Caselli et al. \(2021\)](#) - HateBERT: Retraining BERT for Abusive Language Detection in English.
3. [ElSherief et al. \(2021\)](#) - Latent Hatred: A Benchmark for Understanding Implicit Hate Speech.
4. [Sap et al. \(2020\)](#) - Social Bias Frames: Reasoning about Social and Power Implications of Language
5. [HateXplain dataset](#) - Hate-speech-CNERG/hatexplain, Hugging Face Datasets.